

The Generalization Gap in High-Density Surface-EMG Gesture Decoding: Why Laboratory Accuracy Does Not Survive Contact with the Arm

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Date: June 18, 2026

Document type: Mini-review

Abstract

High-density surface electromyography (HD-sEMG) has made hand-gesture decoding one of the most active problems in non-invasive neural interfacing. Deep-learning classifiers now report offline recognition accuracies above 95%, and in some benchmarks approach 99%. Yet myoelectric prostheses and gesture interfaces remain difficult to use reliably outside the laboratory. This review argues that the field's central obstacle is not raw classification accuracy but **generalization**: the failure of decoders to maintain performance across electrode shift, limb position, recording sessions, and users, compounded by a weak and sometimes absent correlation between offline accuracy and real-time control. I summarize the standard HD-sEMG decoding pipeline, organize the dominant sources of degradation, survey the main mitigation strategies (robust feature/electrode design, transfer learning and domain adaptation, and motor-unit decomposition), and identify open problems. I conclude that progress should be measured by online, cross-condition robustness rather than by offline benchmark numbers, and I briefly situate my own ongoing armband project within this framing.

1. Introduction

Surface electromyography (sEMG) records the electrical activity that muscle fibers generate when motor units fire. Because that activity precedes and accompanies movement, sEMG offers an attractive, non-invasive channel for inferring a user's intended action, most prominently hand gestures for prosthetic control and human-machine interfaces [1]. Conventional myoelectric control uses a small number of electrodes; **high-density** surface EMG instead places tens to hundreds of closely spaced electrodes over a muscle region, capturing the spatial distribution of activity rather than a few isolated channels [2, 15]. This spatial richness is what makes HD-sEMG appealing: it turns a handful of time series into something closer to a spatiotemporal "image" of muscle activity that modern machine-learning models can exploit [3].

The results, measured offline, are striking. Geng and colleagues showed that gesture identity is recoverable even from a *single instantaneous frame* of HD-sEMG, reaching 89.3% accuracy on an 8-gesture task from one frame and 99.0% with majority voting over 40 frames, while also outperforming prior methods on the 52-gesture NinaPro and 27-gesture CSL-HDEMG benchmarks [3]. Subsequent work with 3D convolutional networks, vision transformers, and other deep architectures has pushed within-session accuracy higher still [2, 4].

These numbers invite a natural conclusion, that gesture decoding is essentially solved, which the clinical and engineering reality contradicts. Commercially and clinically, multifunction myoelectric control remains frustrating to use, and abandonment of upper-limb prostheses is well documented, with reported device rejection rates ranging from roughly 23% to 45% depending on device type and population [16]. The gap between benchmark performance and usable control is the subject of this review. I argue that the limiting factor is **generalization across the conditions that change between training and use**, and that the field's evaluation habits (reporting offline, within-session accuracy) actively obscure the problem. This concern is sharpened by the field's broader move toward naturalistic, out-of-laboratory recording: mobile brain-body imaging now captures EEG together with EMG and full-body kinematics during unconstrained locomotion across level ground, ramps, and stairs [17], exactly the kind of changing real-world conditions under which fixed-condition decoders are least tested. The remainder of the paper develops this claim and reviews how researchers are attacking it.

2. The standard HD-sEMG decoding pipeline

Most HD-sEMG gesture decoders share a common structure, which is worth stating explicitly because each stage is also a point where generalization can break down.

Acquisition. An electrode grid (for example, the 8×16 array used in the CapgMyo database, or the 168-channel CSL-HDEMG setup sampled at 2048 Hz) records multi-channel signals during a set of gestures [15]. Public benchmark datasets (NinaPro, CapgMyo, CSL-HDEMG, and others) have standardized this stage and enabled comparison across methods [3, 15].

Feature extraction. Classical myoelectric control relies on hand-crafted features. The most enduring is the Hudgins time-domain set (mean absolute value, waveform length, zero crossings, and slope sign changes), introduced in 1993 and still used as a baseline more than three decades later [5]. These features are computed over short sliding windows to preserve the structure of the contraction.

Classification / decoding. Early systems used linear discriminant analysis (LDA) or neural networks on the extracted features [5]. HD-sEMG, by contrast, increasingly feeds the raw spatial signal directly into deep networks (3D CNNs that jointly capture spatial and temporal structure [2], transformer-based models [4], and recurrent or hybrid architectures) that learn their own features from the electrode grid [3, 4].

Control. In a deployed system the decoded intent drives an actuator (a prosthetic hand, a robotic arm, or a cursor). Critically, this stage introduces a *closed loop*: the user sees the result and adapts, which, as Section 4 discusses, changes the relationship between classifier accuracy and usable performance.

Every one of these stages is typically optimized and reported under fixed conditions: one session, one limb position, one electrode placement, often one subject. The generalization problem is what happens when those conditions move.

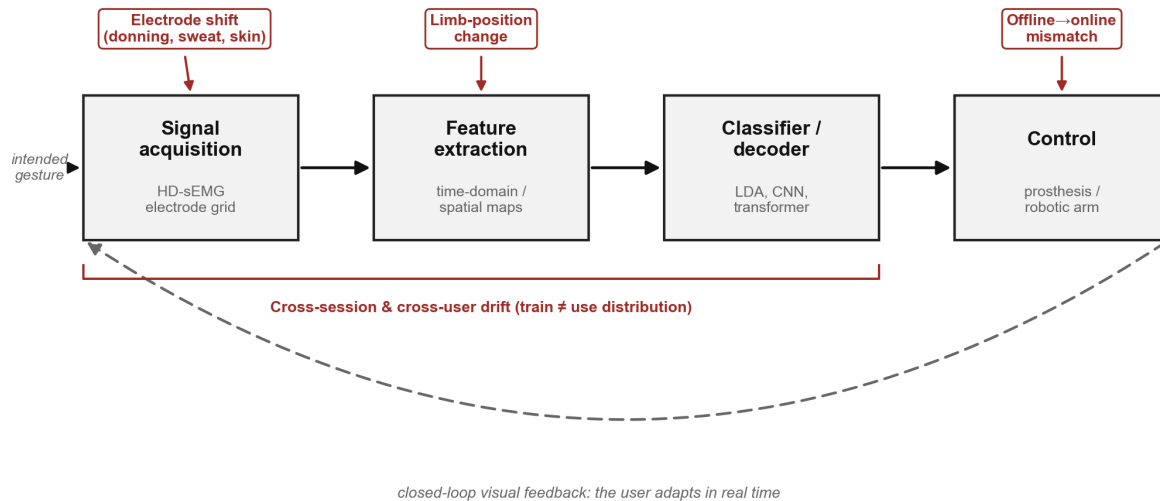


Figure 1. The standard HD-sEMG decoding pipeline (black) and the four points at which laboratory accuracy leaks out before reaching the arm (red). Electrode shift and limb-position change corrupt the signal at acquisition; cross-session and cross-user drift make the training distribution differ from the use distribution across the whole front end; and the offline-to-online mismatch arises at the control stage, where closed-loop feedback lets the user partly compensate, decoupling online control quality from offline accuracy.

3. The generalization gap

Figure 1 maps the four dominant failure modes onto the pipeline; the subsections below take them in turn.

3.1 Electrode shift

Electrodes do not stay where they were during training. Donning and doffing a prosthesis, skin movement, and sweat all displace the grid relative to the muscle. Because HD-sEMG decoders learn spatial patterns, even small shifts change the input distribution. Studies quantifying this effect find substantial accuracy loss, and show that electrode density and configuration modulate the severity: larger inter-electrode distances and combined longitudinal/transverse channel layouts reduce sensitivity to shift [8, 9]. In other words, shift sensitivity is partly a design variable, not only an algorithmic one.

3.2 Limb position

A decoder trained with the arm in one posture degrades when the arm moves to another. Different limb positions recruit muscles differently and change the geometry of muscle, skin, and electrode, producing different signal patterns for the *same intended gesture*. A recent systematic study found that limb position and electrode shift each degrade classification by roughly 14–21%, with no meaningful difference between the two factors; they are comparably damaging [7]. Inter-position classification (train in one posture, test in another) showed average error rising from about 3.8% within a position to roughly 21% across positions [7].

3.3 Cross-session and cross-user variability

Signals drift between recording sessions even for the same person and same electrode placement, and they differ far more between people because of anatomy, skin properties, and motor strategy [10, 11]. A decoder that is excellent within the session it was trained on can fall off sharply on a later session or a new user. This is the variability that domain-adaptation research (Section 5.2) specifically targets, and the reason inter-session and inter-subject evaluation has become a standard, harder benchmark [6, 11].

3.4 The offline–online mismatch

The most consequential and least intuitive part of the gap is that **offline accuracy is a poor predictor of real-time control quality**. In one direct comparison, algorithms achieving roughly 97–98% offline classification accuracy delivered only about 70% accuracy and 71–72% task-completion in real-time use [12]. Strikingly, the relationship can even invert: arm-position change significantly degrades *offline* prediction yet has little effect on *online* control, because the closed-loop user compensates using real-time feedback, so offline and online performance show no significant correlation [12, 13]. The practical implication is severe: a method that wins on a benchmark may not produce a better device, and ranking methods by offline accuracy can be actively misleading.

4. Why the gap persists: evaluation as part of the problem

The four failure modes above are usually studied in isolation, but they share a root cause in how the field measures success. Offline, within-session, within-subject accuracy is cheap to compute, comparable across papers, and steadily improving, which makes it the default headline number. But Section 3.4 shows it does not track the quantity that matters, usable closed-loop control [12, 13]. This creates a perverse incentive: models are tuned to a metric that is partially decoupled from deployment.

A more honest evaluation regime would (i) report cross-session and cross-user results by default rather than as an afterthought [6, 11], (ii) include online, task-based metrics such as completion rate and throughput [12], and (iii) test explicitly under electrode shift and limb-position change rather than holding them fixed [7, 8]. Several of the works reviewed here already do this, and they are the ones whose conclusions are most likely to transfer to a real arm.

5. Approaches to closing the gap

5.1 Robust acquisition and feature design

The earliest and most direct lever is hardware and signal design. Choosing inter-electrode distance and channel orientation to reduce shift sensitivity [9], increasing electrode density where it improves robustness [8], and selecting feature sets known to be more shift-tolerant are all interventions at acquisition/feature time rather than at the model. Their appeal is that they reduce the distribution shift the model must absorb in the first place, rather than asking the model to be invariant to arbitrary change.

5.2 Transfer learning and domain adaptation

The dominant machine-learning response is to treat each new session, posture, or user as a new but related *domain* and adapt to it. Du and colleagues introduced deep domain adaptation for inter-session HD-sEMG recognition, validating on NinaPro, CSL-HDEMG, and CapgMyo and improving cross-session performance over prior methods [6]. Later work has continued this line with multi-scale CNNs pre-trained for inter-subject generalization (using gradient-reversal domain adaptation and self-supervised objectives) [10], lightweight convolutional models that target both inter-session and inter-subject variability [10], and online cross-session pipelines combining deep learning with transfer learning [11]. Reported gains from domain adaptation can be large (on the order of 20–30% over no adaptation in some settings [11]), though, per Section 3.4, such offline gains should be interpreted with caution until confirmed online.

5.3 Motor-unit decomposition: decoding neural drive instead of patterns

A more fundamental shift is to stop classifying signal patterns and instead decompose HD-sEMG into the discharge timings of individual motor units, recovering an estimate of the **neural drive** the spinal cord sends to the muscle. Recent methods can extract on the order of dozens of motor units from a single HD-sEMG array, far more than earlier non-invasive techniques [14]. Because neural drive is closer to the user's intent than the raw interference signal, and because the association between decoded motor-unit activity and force grows stronger as more units are included [14], this approach is a promising route to interfaces that are more interpretable and potentially more robust to the confounds in Section 3. It is also less mature and more computationally demanding than pattern recognition, and its robustness across the same shift/position/session conditions remains an open question.

6. Open problems and where I am headed

Several questions follow directly from this review:

1. **Robustness as the primary metric.** Can the field converge on a shared cross-condition, online benchmark so that methods are compared on what deployment actually requires [7, 12]?
2. **Adaptation with minimal recalibration.** How little new data can a transfer-learning system need to recover performance on a new session or user, and can that recalibration be made unobtrusive to the wearer [6, 10, 11]?
3. **Does neural drive generalize better?** Motor-unit decomposition is more interpretable in principle [14], but it has not yet been systematically stress-tested against electrode shift, limb position, and cross-session drift the way pattern-recognition pipelines have [7, 8].

These questions motivate my own ongoing work: a multi-channel HD surface-EMG armband for gesture decoding, developed alongside a hands-on EMG-controlled robotic arm that serves as a real actuation testbed. The system is at the build-and-instrumentation stage; I am not reporting decoding results here, and the project is described only to indicate the applied direction in which I am reading this literature. My intent is to design data collection around the generalization conditions above from the start, varying limb position and session deliberately and evaluating online rather than only offline, so that whatever accuracy the system reaches is accuracy of the kind that survives contact with the

arm.

7. Conclusion

HD-sEMG gesture decoding has a deceptive scoreboard. Offline, within-session accuracy has climbed toward ceiling, but this number understates the difficulty of building a device a person can rely on. The binding constraints are electrode shift, limb-position change, cross-session and cross-user drift, and the weak, sometimes inverted relationship between offline accuracy and online control. The most credible research either reduces the distribution shift at the source, adapts across domains, or moves toward decoding neural drive and, crucially, evaluates under the conditions of real use. Treating cross-condition, online robustness as the primary objective, rather than offline benchmark accuracy, is the change most likely to turn impressive results into usable interfaces.

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Note on scope and honesty: This is a literature review written by a high-school student preparing for undergraduate work in biomedical/neural engineering. It reports no original experimental data. All cited works are real, peer-reviewed (or archival preprints where noted), and listed with DOIs/links. Quantitative figures (e.g., the 14–21% degradation [7] and the ~97% offline vs. ~70% online accuracy [12]) are drawn from and attributed to the cited papers.